



Linda S. Adams
Acting Secretary for
Environmental Protection

California Regional Water Quality Control Board Central Valley Region

Katherine Hart, Chair

415 Knollcrest Drive, Suite 100, Redding, California 96002
(530) 224-4845 • FAX (530) 224-4857
<http://www.waterboards.ca.gov/centralvalley>



Edmund G. Brown Jr.
Governor

25 April 2011

WDID 5A520101001

Mr. Mark Barthel, Director Public Works
City of Red Bluff
555 Washington Street
Red Bluff, CA 96080

INCOMPLETE APPLICATION/REPORT OF WASTE DISCHARGE, RED BLUFF WASTEWATER RECLAMATION PLANT, WASTE DISCHARGE REQUIREMENTS ORDER NO. R5-2007-0041, NPDES PERMIT NO. CA0078891, TEHAMA COUNTY

Central Valley Regional Water Quality Control Board (Central Valley Water Board) staff has reviewed the 1 April 2011 renewal *Application/Report of Waste Discharge for NPDES Permit-City of Red Bluff Wastewater Reclamation Plant – Tehama County*, prepared by SouthWest Water Company. The application is hereby deemed **incomplete**. The following additional information is necessary for renewal of Waste Discharge Requirements, which expire on 1 May 2012:

- Signature of authorized official: The renewal application/ report of waste discharge must be signed by an appropriate representative of the City of Red Bluff, for example the public works director;
- Mixing Zone/Dilution Credits: The current NPDES permit includes dilution credits of 26:1 for chronic, acute, and human health criteria for certain water quality pollutants. The following information should be provided to the Central Valley Water Board if the City of Red Bluff is requesting the use of dilution credits in the calculation of water quality-based effluent limits:
 1. Up to date field dye study and/or mixing zone modeling results,
 2. Biological assessment of the proposed mixing zone,
 3. California Department of Fish and Game (DFG) review of the proposed mixing zone, and
 4. Any additional pertinent water quality data.

Additionally, Central Valley Water Board staff has completed a case file review and an initial review of the electronic monitoring data available in the California Integrated Water Quality System (CIWQS) for the Red Bluff Wastewater Reclamation Plant in preparation for the upcoming NPDES permit renewal. The following is a summary of pertinent information regarding the subject renewal.

Chronic Toxicity

The current NPDES permit contains a numeric chronic toxicity monitoring trigger of greater than 10 (>10). It is anticipated that the draft NPDES permit will include a toxicity

monitoring trigger of 1. If the discharger feels a trigger greater than 1 is appropriate, the following must be provided:

- Justification that a higher monitoring trigger is appropriate;
- Justification of appropriate dilution credit and size of the mixing zone;
- Justification that mixing zone conditions outlined in State Implementation Policy (SIP) Section 1.4.2.2 will be met; and
- Letter specifically requesting dilution credits for chronic toxicity.

pH

The current NPDES permit contains effluent limits for pH between 6.0 and 9.0. It is anticipated that the draft NPDES permit will include effluent limitations for pH between 6.5 and 8.5 at the water quality objectives. If the discharger feels that the range of limitation pH should be outside the water quality objective, justification must be provided.

Mixing Zone/Dilution Credits

In 2006, the City of Red Bluff conducted a Mixing Zone Study for the discharge of wastewater to the Sacramento River from the Red Bluff Wastewater Reclamation Plant. At the time the Red Bluff Diversion Dam (RBDD) was operational. Between May 15th and September 15th the RBDD gates were lowered into the Sacramento River, allowing water to be diverted by gravity flow into the Corning and Tehama-Colusa irrigation canals. Because fish passage was difficult while the RBDD gates were lowered, a new pumping plant was constructed to divert irrigation water and the use of the RBDD was discontinued. The RBDD gates will now be raised year round. Conditions corresponding to operation of the diversion dam were reflected in the 2006 mixing zone study where it showed that the critical condition for dilution is when the diversion dam is open. When the RBDD was closed, the elevation in the river was approximately 5 feet higher than during free flow periods. As a result of discontinued use of the diversion dam, river conditions may have changed. Additionally, Sacramento River flow measurements at the Big Bend USGS station #11377100 show that the minimum river flows have decreased in recent years, since at least 2006 to present. These changing conditions indicate that assimilative capacity in the river needs to be reexamined if the discharger is requesting dilution credits.

Dilution Credits

The current NPDES permit includes dilution credits of 26:1 for chronic, acute, and human health criteria. Water quality based effluent limitations for constituents showing reasonable potential to affect water quality including, copper, zinc, chlorodibromomethane, and dibromochloromethane, were calculated based on the dilution credits. Water quality data since 2007, show that the following constituents have reasonable potential to affect water quality: copper, zinc, chloroform, dibromochloromethane, and ammonia. The draft NPDES permit may include effluent limits for these pollutants.

It is anticipated that the draft NPDES permit will include a requirement to perform a revised mixing zone and tracer study/dye test to determine mixing zone/dilution conditions in the Sacramento River below the discharge location.

25 April 2010

The draft NPDES permit will include a time schedule to allow the discharger to complete a new mixing zone study.

Your complete Application/Report of Waste Discharge, which includes the items described above must be filed with the Central Valley Water Board **by 1 November 2011**.

If you have any additional questions, please contact Katie Bowman of my staff at (530) 226-3458, kbowman@waterboards.ca.gov, or the letterhead address.

(original signed by Bryan Smith)

Gregory D. Cash
Senior Engineering Geologist
South Regulatory Unit

KB: lm

cc: Ms. Kathy Stone, SWWC, Corning